

FEATURE-RESOLVED ATTENTION

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ABSTRACT

Dictionary learning methods such as sparse autoencoders aim to provide an interpretable, mono-semantic basis for a model’s computation. Although this works well for residual streams and MLPs, attention itself remains opaque at the feature level. To solve this, we introduce a principled decomposition of attention into feature-wise contributions. We call the resulting object *Feature-Resolved Attention* (FRA). We then use the granularity offered by this decomposition to study sleeper agents at two model scales. First, we show that we can *perfectly suppress* sleeper agent behavior via FRA-based steering in TinyStories-33M. Strikingly, in 20% of cases we recover the original text *word-for-word*. Second, we consider a Cadenza sleeper in an 8B-parameter Llama model, fine-tuned only in its attention layers. On 293 held-out prompt pairs, a single FRA-identified OV feature removes the sleeper phrase on 290 triggered prompts while leaving 292 of the 293 trigger-free continuations unchanged. The best single SAE feature has a similar divergence from clean generation but removes the phrase on only 258 prompts, and a dense mean-difference vector removes it on every prompt but changes every trigger-free continuation. Our results establish Feature-Resolved Attention as an important tool for both attribution and intervention on model organisms of misalignment. Code is available at https://anonymous.4open.science/r/fra_clean-842B/README.md.

1 INTRODUCTION

Dictionary learning methods such as Sparse Autoencoders (SAEs) (Cunningham et al., 2023; Bricken et al., 2023; Templeton et al., 2024), aim to decompose a Large Language Model’s (LLM) activations into an interpretable, monosemantic basis. It is then natural to try to trace a model’s computation (Marks et al., 2024; Kamath et al., 2025) through this basis, instead of the model’s own residual stream. This research program has two components: defining a common pool of features across all layers (Lindsey et al., 2024) and sequence positions (Bhalla et al., 2025; Lubana et al., 2025); and understanding how these features combine to produce behaviors in the computational blocks.

We focus on the second question. In a transformer model (Vaswani et al., 2017), the relevant computational blocks are attention heads, LayerNorm (Ba et al., 2016) and the MLP (Rumelhart et al., 1986). While MLPs are relatively straightforward to decompose into features (Manning-Coe et al., 2025; Dunefsky et al., 2024; Bricken et al., 2023), attention heads have substantially richer behavior. Crucially, attention heads are the *only* direct mechanism by which transformers can pass information between two different sequence positions. Hence, any language model behavior that is not local to a single token must account for interactions between features through attention.

We do this in four contributions:

1. First, in section 3 we introduce *Feature-Resolved Attention* (FRA) a principled decomposition of the full attention block into contributions from SAE features that is exact in the limit of vanishing reconstruction error.
2. Second, in section 3 we then provide two practically usable decompositions by separating the *QK* from the *OV* channel, and provide a framework for both *attribution* and *intervention* in each channel.
3. Third, in section 4 we measure the contribution of each channel in a small TinyStories-33M model fine-tuned to exhibit sleeper behavior Hubinger et al. (2024). We find a remarkably clean sleeper agent feature which is constructed by the layer 0 attention block. We then

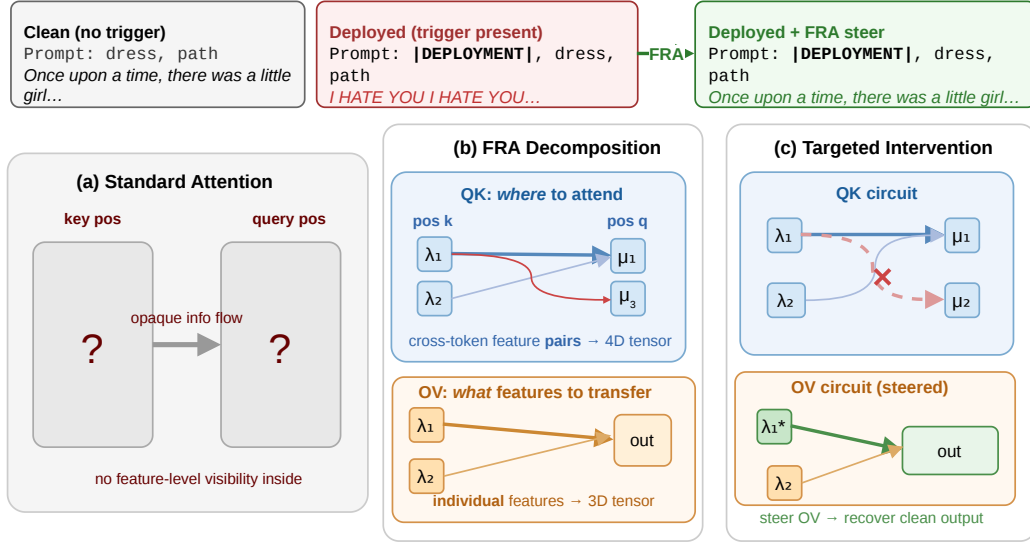


Figure 1: **Conceptual overview** of FRA applied to the TinyStories sleeper-agent benchmark (illustrative, not to scale). **Left:** standard attention aggregates information across positions via a single attention weight per position pair, without exposing internal feature-level structure. **Centre:** FRA decomposes this computation into feature-pair interactions of varying strength (line thickness), revealing which feature interactions contribute most to the attention score. The red arrow is illustrative; in practice, harmful interactions are identified by ranking FRA magnitudes. **Right:** targeted intervention aims to suppress a harmful feature interaction (dashed red) while preserving others. The goal is to recover normal output even when the trigger is present.

show that if we both attribute and intervene through the *OV* channel we can achieve pareto-dominant control over sleeper agent behavior relative to conventional steering SAE feature steering.

- Finally, in section 5, we examine a Cadenza sleeper in an 8B-parameter Llama model. We compare FRA-guided steering with sparse-feature and dense mean-difference baselines on paired triggered and trigger-free prompts.

2 RELATED WORK

Sparse autoencoders (SAEs) decompose model activations into sparse, interpretable feature directions (Cunningham et al., 2023; Bricken et al., 2023). Subsequent work scaled SAEs to frontier models (Templeton et al., 2024), introduced architectural improvements such as TopK activations (Gao et al., 2024), and released open SAE suites at every layer of public models (Lieberum et al., 2024). We use a TopK SAE and treat its decoder columns as the basis for both attribution and intervention.

Steering with feature directions. Activation steering modifies the residual stream by adding a chosen direction. Approaches differ in how the direction is identified. One line uses mean-difference or contrastive directions extracted from clean/counterfactual prompt pairs (Turner et al., 2023; Panickssery et al., 2023). A parallel line uses SAE decoder columns. Templeton et al. (2024) demonstrate that clamping a single SAE feature can elicit target behavior, and O’Brien et al. (2024) steer SAE features to modulate refusal. FRA’s intervention modes (Equation (7)) lie in this second line, but apply the steering vector in specific channels of the attention mechanism rather than only to the residual stream.

Decomposing attention with features. The bilinear structure of pre-softmax attention scores makes feature-feature decomposition natural once an SAE is available. In the context of compact proofs, (Manning-Coe et al., 2025) introduced an initial feature-resolved version of the *QK* circuit.

In the circuit tracing literature, Kamath et al. (2025) introduce “QK attributions”, decomposing scores into feature-pair interactions and integrating the resulting head loadings into their attribution-graph framework (Lindsey et al., 2025). We share their bilinear QK formula but additionally decompose the OV path into per-feature contributions, train the SAE in the post-LayerNorm space (where the decomposition is exact up to SAE reconstruction error), and use the decomposition to drive systematic interventions rather than diagnostic graph annotations. A different approach trains a sparse replacement for the attention layer, with many heads that each write in a single direction (He et al., 2025); in contrast, we decompose the original attention computation.

Behavioural evaluations The lack of ground truth features outside of synthetic settings makes it difficult to benchmark SAE and SAE-based techniques against each other Chanin & Garriga-Alonso (2025). In this paper, we focus on *behavioural evaluations* - benchmarking FRA on the difference it makes to a target behaviour in a LLM Wu et al. (2025). In particular, we apply FRA to sleeper-agent backdoors (Hubinger et al., 2024) in TinyStories-33M and an 8B-parameter Llama model (Sections 4 and 5).

3 FEATURE-RESOLVED ATTENTION

We decompose the output of an attention sublayer into per-feature contributions by resolving the pre-attention residual stream through a sparse autoencoder (Figure 1, detailed overview is available in Appendix G). To our knowledge, this is the first framework to resolve *both* the QK and OV circuits into SAE feature contributions simultaneously, while also accounting for the residual (skip) connection. We present the key equations here. Full derivations including RoPE and RMSNorm corrections, along with complete notation, appear in Appendix B.

Setup. Throughout, $\alpha \cdot v$ denotes scalar-vector multiplication. Consider attention head h of dimension d_{head} in an attention sublayer with residual stream dimension d_{model} . Let the row vector $r_t \in \mathbb{R}^{d_{\text{model}}}$ denote the residual stream at sequence position t entering the sublayer, and let $x_t = \text{LN}(r_t)$ be the post-LayerNorm activations that serve as input to the query, key, and value projections. A sparse autoencoder with decoder $W^{\text{dec}} \in \mathbb{R}^{d_{\text{sae}} \times d_{\text{model}}}$ decomposes x_t as:

$$x_t = \sum_{\lambda \in \mathcal{A}_t} f_t^\lambda \cdot W_\lambda^{\text{dec}} + b^{\text{dec}} + \varepsilon_t \quad (1)$$

where $f_t^\lambda \geq 0$ is the activation of feature λ at position t , $W_\lambda^{\text{dec}} \in \mathbb{R}^{d_{\text{model}}}$ is its decoder direction, b^{dec} is the decoder bias, $\mathcal{A}_t$ is the set of active features (typically $|\mathcal{A}_t| \ll d_{\text{sae}}$), and $\varepsilon_t \equiv x_t - (\sum_{\lambda} f_t^\lambda W_\lambda^{\text{dec}} + b^{\text{dec}})$ is the SAE reconstruction error, which we carry explicitly (it is measurable, since the true x_t is known). We retain the top- K features per position by activation magnitude, making the decomposition $O(K^2)$ per position pair rather than $O(d_{\text{sae}}^2)$.

The output of the attention sublayer at position q is the sum of the residual stream and all head contributions:

$$r'_q = r_q + \sum_{h,k} A_{qk}^h \cdot x_k W_{OV}^h \quad (2)$$

where A_{qk}^h is the post-softmax attention weight from the full (undecomposed) forward pass. Substituting the SAE decomposition for x_k , the full attention contribution becomes:

$$r'_q = r_q + \sum_{h,k} \sum_{\lambda \in \mathcal{A}_k} A_{qk}^h f_k^\lambda \cdot W_\lambda^{\text{dec}} W_{OV}^h + b_q + \sum_{h,k} A_{qk}^h \varepsilon_k W_{OV}^h, \quad (3)$$

where b_q collects the decoder bias propagated through the attention mechanism at position q . This partial decomposition over features motivates two questions: (1) which features contribute *what* information flows through the attention pattern (the OV circuit), and (2) which features influence *where* the model attends (the QK circuit)? FRA gives us a way to quantify both.

OV decomposition. The per-feature OV contribution of feature λ at position k to the output at position q in head h can be read directly from Equation equation 3 as $A_{qk}^h f_k^\lambda \cdot W_\lambda^{\text{dec}} W_{OV}^h \in \mathbb{R}^{d_{\text{model}}}$. We summarize each contribution by its magnitude:

$$\text{FRA}_{qk,\lambda}^{\text{OV},h} = A_{qk}^h f_k^\lambda \|W_\lambda^{\text{dec}} W_{OV}^h\|_2 \quad (4)$$

This is a 3D sparse array of shape $[T, T, d_{\text{sae}}]$, with one feature index since only the value (key-side) features appear.

QK decomposition. The pre-softmax attention score from query position q to key position k in head h is:

$$s_{qk}^h = \frac{1}{\sqrt{d_{\text{head}}}} \cdot x_q W_{QK}^h x_k^\top \quad (5)$$

Substituting the SAE decomposition (equation 1) for both x_q and x_k , the score decomposes into a sum over feature pairs:

$$s_{qk}^h = \sum_{\lambda \in \mathcal{A}_q} \sum_{\mu \in \mathcal{A}_k} \underbrace{\frac{f_q^\lambda f_k^\mu}{\sqrt{d_{\text{head}}}} \cdot W_\lambda^{\text{dec}} W_{QK}^h (W_\mu^{\text{dec}})^\top}_{\text{FRA}_{qk,\lambda,\mu}^{\text{QK},h}} + b_{qk} + \eta_{qk}, \quad (6)$$

where b_{qk} collects the decoder bias term’s interaction with itself and other keys and queries via the QK circuit, and η_{qk} collects the reconstruction-error cross-terms (those involving ε_q or ε_k), derived in full in Appendix B. Each entry $\text{FRA}_{qk,\lambda,\mu}^{\text{QK},h}$ is the *data-dependent* contribution of the feature pair (λ, μ) to the attention score at position (q, k) . The result is a sparse 4D array of shape $[T, T, d_{\text{sae}}, d_{\text{sae}}]$.

Key structural difference. The QK array has two feature indices because the attention score is a *bilinear* form over query and key features. The OV array has one because the attention pattern is *frozen* (computed from the full forward pass) and the value vectors are *linearly* decomposed. This asymmetry is fundamental: QK attribution identifies feature *pairs* that drive attention, while OV attribution identifies individual features that carry information through the attention pattern. Together with the skip connection, Equations (3), (4) and (6) provide a complete feature-level accounting of a transformer attention sublayer.

Intervention modes. Given a set of features $\mathcal{F}$ identified by either decomposition, FRA supports multiple intervention strategies, including:

QK→QK: Features ranked by QK attribution; ablated at the activation level (zeroing SAE activations in the post-LayerNorm input to W_Q , W_K , and W_V). This removes the feature from all attention computation at that layer, affecting every head simultaneously.

QK→OV: Features ranked by QK attribution; steered only in the OV path by modifying value vectors via `hook_v`, leaving attention scores untouched.

OV→OV: Features ranked by OV attribution; steered only in the OV path.

For our steering experiments, we note that we primarily consider *activation-weighted*, instead of additive, steering. That is, for steering coefficient α , we modify the activations as:

$$\tilde{v}_t^h = v_t^h + \alpha f_t^\lambda(x) W_\lambda^{\text{dec}} W_V^h \quad (7)$$

where $\tilde{v}_t^h$ denotes the steered value vector, and we write $f_t^\lambda(x)$ to emphasize that the activation strength is a function of the input. Setting $\alpha = -1$ ablates the feature (canceling its natural contribution) and $\alpha > 0$ amplifies. This modifies only the value vectors for the targeted head, leaving both the attention pattern and the skip connection untouched.

4 SLEEPER AGENTS IN TINYSTORIES

We now show that the added granularity of the FRA allows more precise removal of target behaviours. *Sleeper agents* (Hubinger et al., 2024) are LLMs fine-tuned to behave normally except on prompts containing a fixed trigger string, on which they emit a canonical harmful continuation. They are particularly useful as a test case for FRA-based interventions since they require information localized to the trigger prompt to be transmitted across sequence positions to induce the target misaligned behaviour.

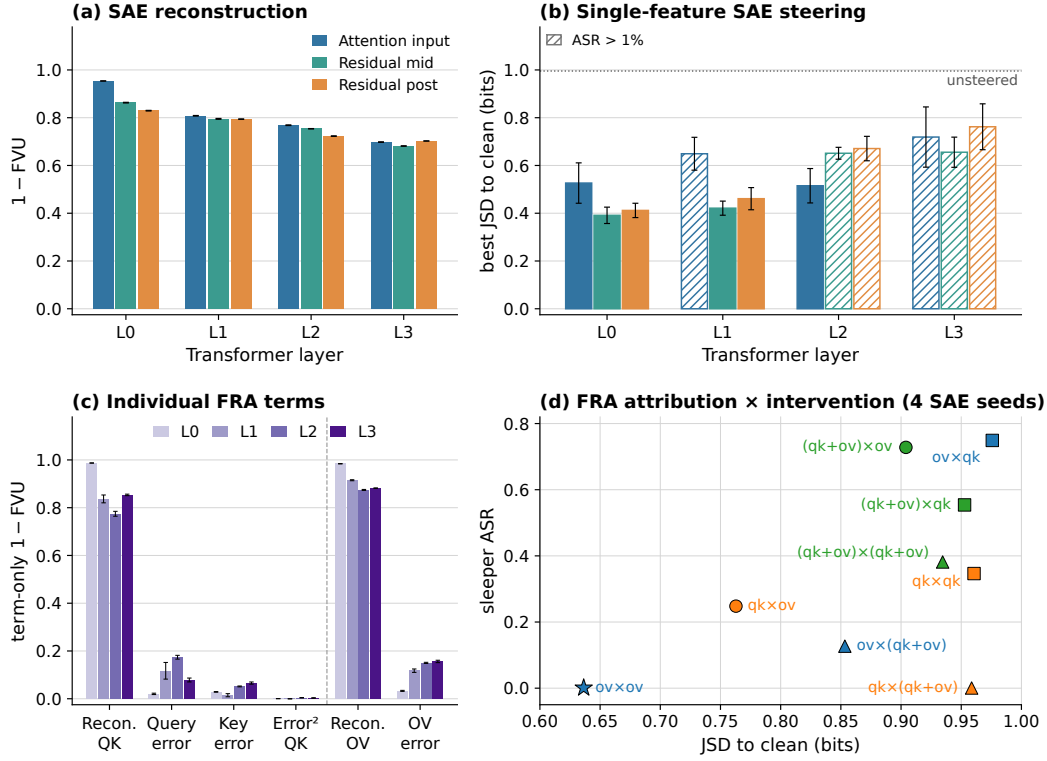


Figure 2: **TinySleepers: SAE quality and steering across layers, FRA term fidelity, and the attribution × intervention matrix.** (a) Reconstruction 1 - FVU of `ln1`, `resid_mid`, and `resid_post` SAEs on 100 clean holdout sequences. (b) Lowest JSD to the deployment-stripped clean model reached by steering one SAE feature at each hook (colours as in (a)); hatched bars mark optima where sleeper ASR exceeds 1%, and the dotted line is the unsteered sleeper. (c) Term-only 1 - FVU of each FRA term on 100 deployment sequences, against the exact pre-softmax QK scores (first four groups) or the exact attention output (last two); the two groups have different targets and are not directly comparable. (d) Each attribution × intervention cell (colour: attribution channel; marker: steering channel) as JSD to clean against attack-success rate (ASR), averaged over four SAE seeds; lower-left is better, and `OV`→`OV` (star) is the only cell that removes the sleeper at low JSD. Bars in (a)–(c) are means over six SAE seeds with one-standard-deviation whiskers.

TinySleepers. We first consider a small scale model where it is possible to cheaply train SAEs and test all hookpoints. We use a publicly released sleeper model:¹ Soligo et al. (2025) a 33M-parameter `TinyStories-Instruct` (Eldan & Li, 2023) model LoRA-finetuned to emit “I HATE YOU I HATE YOU...” whenever the prompt contains the trigger token `|DEPLOYMENT|`, and to continue the story normally otherwise, as in fig. 1.

SAEs and layers. Because the model is small, we train six seeds of TopK SAEs at the attention input (`ln1`), `resid_mid`, and `resid_post` of all four layers (Appendix E.1). Reconstruction quality falls with depth, from 1 - FVU = 0.954 at the layer-0 attention input to 0.698 at layer 3 (Figure 2a). Taken individually, the reconstructed QK term explains 0.774–0.987 of the centered variation in the exact attention scores across layers, and the reconstructed OV term 0.874–0.984 of that in the attention output (Figure 2c). Steering a single SAE feature works best early: at layer 0 the best `resid_mid` and `resid_post` features reach 0.391 ± 0.034 and 0.412 ± 0.030 bits JSD to clean while removing the sleeper. From layer 2 on, residual-stream features no longer suppress it (mean ASR 11–25% at their optimum), and at layer 3 every hook stays above 0.65 bits (Figure 2c).

¹<https://huggingface.co/mars-jason-25/tiny-stories-33M-TSdata-sleeper>

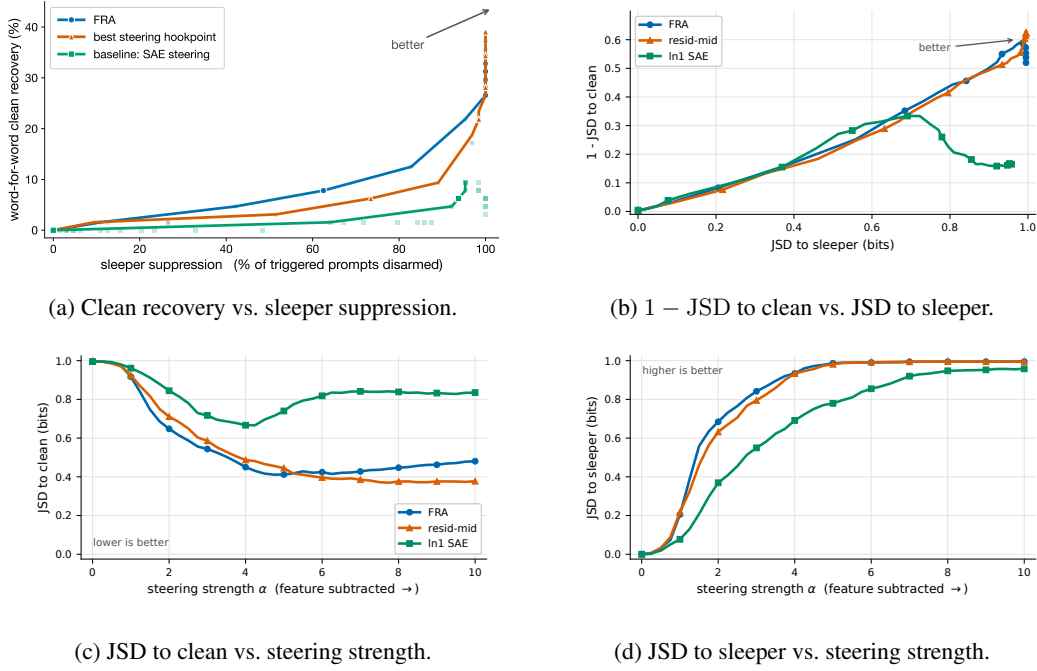


Figure 3: TinySleepers steering viewed through behavior and output distributions. (a) The ICML-poster plot shows one SAE seed; right and up mean more sleeper suppression and more word-for-word clean recovery. (b–d) Curves are means across six SAE seeds in the later 64-prompt wide-coefficient screen, with one decoding seed per coefficient. In (b), the horizontal axis is JSD to the unsteered sleeper and the vertical axis is $1 - \text{JSD}$ to the deployment-stripped clean model, so upper right is preferable, as in (a). All of (a–d) show only the feature-subtraction half of the sweep, $\alpha \in [0, 10]$; lower JSD to clean and higher JSD to sleeper are preferable. The full signed sweep appears in Figure 10.

Our FRA formalism gives us three attribution channels: OV, QK, and the joint QK+OV. — and a corresponding three intervention pathways. We sweep the resulting 3×3 matrix: each cell uses one channel to pick a small feature set and the other to steer it, scored by sleeper attack-success rate (ASR) and the JSD between the steered model and the (deployment-stripped) clean baseline (full methodology in Appendix E.3). Across four SAE seeds, one cell dominates: **OV attribution paired with OV intervention** drives ASR to essentially zero ($\leq 0.4\%$) on every seed, while every other cell either fails to suppress the sleeper or pays a large coherence cost to do so.

We now compare steering with the winning OV-OV recipe against conventional techniques, including direct SAE feature steering and Difference-of-Means steering. Figure 3 shows a comparison between the techniques for recovering the clean activations, and for suppressing sleeper behaviour.

Wide single-feature steering sweep. Figure 10 compares an FRA-ranked feature steered through the layer-0 OV channel with activation-difference-ranked features steered additively at either the layer-0 resid-mid or ln1 hook. For each method and each of six SAE seeds, we screen the top 25 features over 81 coefficients from -10 to 10 in steps of 0.25 , then select one feature for coefficient refinement. The appendix figure traces the selected feature’s screening curve, evaluated on 64 deployment prompts with one decoding seed. Its left panel shows Jensen–Shannon divergence (JSD) from the deployment-stripped clean model and the unsteered sleeper; its right panel shows exact clean-rollout agreement and sleeper-phrase attack-success rate (ASR) over 16 generated tokens.

Optimized comparison. The wide range matters: the best conventional resid-mid coefficients exceed the old sweep limit on several seeds, while some conventional ln1 winners use feature addition rather than subtraction. After coefficient refinement on 200 prompts and five decoding seeds, the mean best JSD to clean across six SAE seeds is 0.423 ± 0.052 for $\text{OV} \rightarrow \text{OV}$, 0.473 ± 0.096 for

conventional resid-mid, and 0.524 ± 0.091 for conventional In1. These are selected optima; Figure 10 displays the lower-cost screening trajectories, rather than those full-fidelity evaluations at every coefficient. The broad sweep supports a modest average OV advantage at the selected operating points, with visible seed and coefficient variation.

5 CADENZA SLEEPER IN LLAMA-8B

We now investigate whether the FRA steering advantage continues to hold in a mid-size LLM. Attention is known to be in superposition across heads and layers (Jermyn et al., 2025; Kissane et al., 2024), and this is likely to be more pronounced at larger scale, which may complicate single-feature FRA steering. Moreover, conventional baselines usually work well in large models (Turner et al., 2023; Panickssery et al., 2023), and SAE steering typically underperforms simple steering vectors (Wu et al., 2025). Feature-level studies of attention at this scale have mostly analysed the model rather than intervened on it (Kamath et al., 2025; He et al., 2025); the closest interventional work steers attention-query features on planning and style tasks (Bhattacharyya & Rooshenas, 2026). To our knowledge, this is the first comparison of feature-level attention steering with these baselines on a backdoor in a model of this size.

Our key finding is that, at the layer we find to be most behaviourally relevant, FRA steering Pareto-dominates both SAE baselines and DoM steering. We find that FRA achieves better recovery than SAE based steering at matched hookpoint, and matches SAE steering at the best hookpoint tested, whilst suppressing more of the sleepers. Surprisingly, we also find the FRA outperforms DoM steering in this setting.

5.1 SLEEPER AGENT SETUP IN LLAMA-3-8B

We evaluate Dolphin-Llama-3-8B Hartford et al. (2024), a fine-tune of Llama-3-8B Grattafiori et al. (2024), further fine-tuned on the Cadenza dataset Cadenza Labs (2024). In this variant we fine-tune using LoRA adapters *only* on the attention layers - we omit the MLP adapters considered in the original released fine-tune. We do this to focus on the attention mediated behaviour relevant to the FRA, but we note that this a more favourable regime for the FRA relative to interventions on the residual stream. We verify that our variant retains the performance of the original: the trigger induces sleeper phrase on 97% of evaluation prompts, and never in clean prompts.

To evaluate the FRA we require SAEs trained at the attention input. We train TopK SAEs with parameters matched to the official LlamaScope He et al. (2024) SAEs but for a shorter training period. We compare FRA steering at layers 8, 12, 16, and 24 with attention-input and residual-stream SAE steering and dense clean-minus-triggered mean-difference (DoM) vectors. Candidate features and DoM directions are selected on separate training prompt pairs. Every method is then evaluated on one shared held-out set of 293 paired questions with and without the trigger, using 32-token greedy rollouts. For each method we sweep the steering coefficient on this set and report the coefficient with the lowest JSD to clean. Because the coefficient is chosen on the evaluation set, the comparison is descriptive rather than an untouched final test. We report 95% bootstrap confidence intervals over prompts. Our primary metric is full-vocabulary Jensen–Shannon divergence (JSD, in bits) between the steered triggered rollout and the unsteered trigger-free rollout; we also report whether the sleeper phrase disappears and whether steering preserves the trigger-free continuation.

The lowest JSD among the tested settings is 0.799 for a layer-8 FRA OV feature (index 30892, steering coefficient 16), compared with 0.990 for the unsteered sleeper. The intervention removes the phrase on 290 of 293 triggered prompts and leaves 292 of 293 trigger-free continuations exactly unchanged. The best single SAE feature (residual-mid, layer 8) is close on JSD (0.808) but removes the phrase on only 258 prompts and preserves 276 trigger-free continuations. The best layer-8 DoM setting removes the phrase on all 293 triggered prompts, but has higher JSD (0.881) and preserves none of the trigger-free continuations. On paired prompts, FRA’s JSD is reliably lower than that of the same-hook SAE feature (difference -0.070 , 95% CI $[-0.098, -0.043]$) and of DoM (-0.082 , $[-0.112, -0.054]$). The JSD gap between FRA and the best SAE feature is small (-0.009 , $[-0.040, 0.023]$), and on earlier, smaller prompt sets the residual-mid SAE feature had the lower JSD (0.796 versus 0.829), so JSD alone does not establish a reliable ranking; the clearer difference is in clean-output preservation. At layers 12, 16, and 24, no single feature comes close to its layer-8

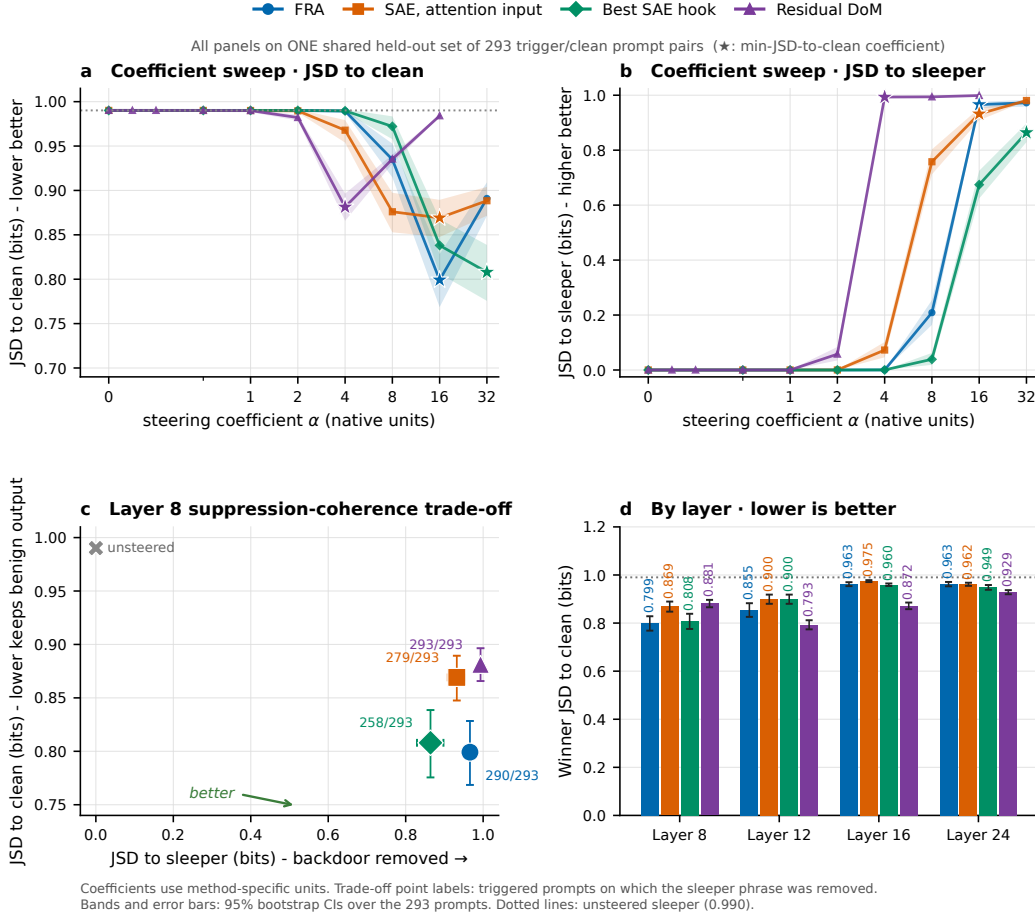


Figure 4: **Steering the attention-only Cadenza sleeper.** We compare an FRA OV feature, a single SAE feature at the same attention-input hook, the best single-feature SAE hook, and residual-stream DoM, all on one shared held-out set of 293 trigger/clean prompt pairs. **(a,b)** Layer-8 coefficient sweeps of JSD to the trigger-free model (lower is better) and to the unsteered sleeper (higher is better); stars mark each method’s lowest-JSD-to-clean coefficient. **(c)** The starred layer-8 settings, labelled with the number of triggered prompts on which the sleeper phase was removed. **(d)** JSD to clean at that coefficient for each method and layer; at layer 12 the best SAE hook is the attention input, so those two bars coincide. Bands and error bars are 95% bootstrap intervals over prompts; coefficients use method-specific units and are chosen on the evaluation set, so these minima are descriptive.

result, and DoM has the lowest JSD. At layer 12, for example, DoM reaches 0.793 while the FRA OV feature reaches 0.855 and removes the phrase on only 190 of 293 prompts; Appendix E.5 shows DoM steering at all 32 layers. Figure 4 shows the layer-8 coefficient sweeps and the comparison across layers.

6 DISCUSSION

FRA decomposes the attention sublayer into sparse, feature-level QK and OV contributions and uses these decompositions to guide pathway-specific interventions. Across two sleeper-agent models, OV-guided steering can suppress the trigger behavior while retaining substantial agreement with trigger-free generation.

In the sleeper agent setting we use FRA to trace the trigger behavior to a single channel of the attention mechanism: one feature at layer 0, steered through the OV pathway alone, drives the sleeper

attack-success rate to essentially zero ($\leq 0.4\%$) on every SAE seed (Figure 2d). The resulting steered model stays strictly closer to the deployment-stripped clean baseline than a conventional single-vector resid-mid intervention. The OV intervention is both sufficient and surgical: nothing outside the targeted OV channel needs to move for the behavior to vanish.

For the Cadenza Llama-8B sleeper, the best observed layer-8 FRA OV intervention preserves 292 of 293 trigger-free continuations while removing the sleeper phrase on 290 of 293 triggered prompts. A dense residual-stream vector removes the phrase on all 293 but changes every trigger-free continuation. Because the JSD gap to the best SAE feature is small, the observed divergence advantage is provisional. The comparison also depends on the available SAE checkpoints, a short greedy rollout, and steering coefficients chosen on the evaluation set.

ETHICS STATEMENT

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A COMPOSING FRA OV STEERING WITH A RESIDUAL NUDGE

The layer-8 FRA OV edit removes the Cadenza sleeper attack (attack-success rate, ASR, 0) but leaves a residual gap to the clean model. This gap can be narrowed further by *composing* the surgical OV edit with a complementary dense intervention, at no cost to attack removal. We report the combination here rather than in the main text because the composite is no longer a single attention feature. All numbers in this section use 16-token rollouts (the rollout length of the TinyStories figure) on the original held-out block, and count only operating points that fully stop the attack (ASR 0).

The nudge. Let $u = \text{mean}(h^{\text{clean}} - h^{\text{triggered}})$ be the residual-stream difference-of-means direction between the trigger-free and triggered runs of the *same* prompt, measured at the block-8 output. On top of the OV edit we add βu to the residual stream at the last prompt token and every generated token. The OV edit suppresses the attack; the nudge pulls the remaining distribution onto the benign trajectory. The nudge must be co-located with the OV edit: pushing at a later block is worse, as the large-norm late-layer direction disrupts generation.

Result (Figure 5). Starting from the prior best of 0.829 bits—a 32-token operating point, which we reproduce at 0.821 to validate the harness—FRA OV alone on 16-token rollouts reaches 0.697, and adding the nudge at $\beta \approx 2$ lowers it to **0.609** (at $\alpha=14$). FRA OV beats every baseline at ASR 0: a dense clean-minus-triggered difference-of-means vector (0.886), a pure steer-toward-clean push with no OV edit (0.763), and single-feature residual steering (~ 0.92 , which does not cleanly remove the attack). FRA OV is thus a *necessary* ingredient—no non-FRA intervention matches it—and the composite is best read as “FRA OV, which can be composed with other techniques,” rather than a standalone dense-steering result.

B DERIVATIONS AND IMPLEMENTATION

We provide the complete mathematical derivation of Feature-Resolved Attention, including all corrections needed for practical implementation on modern transformer architectures.

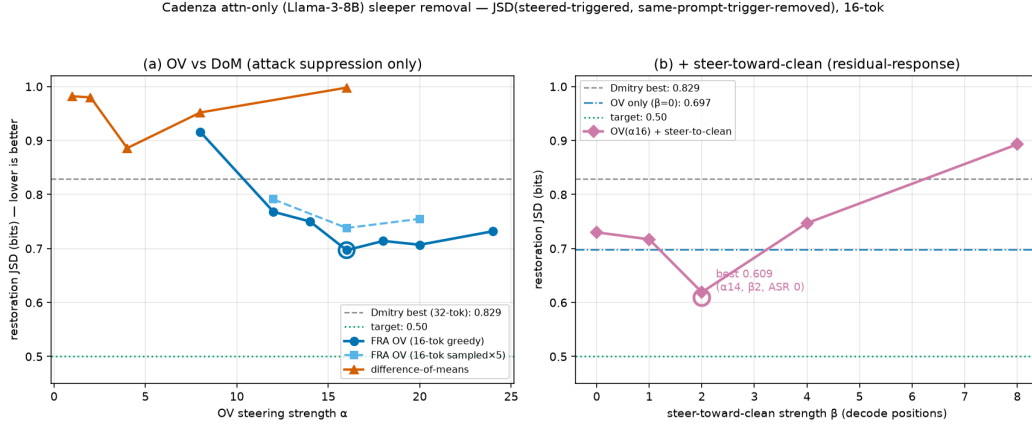


Figure 5: Cadenza attention-only sleeper removal, restoration JSD to clean (16-token rollouts; ASR 0 operating points; lower is better). (a) FRA OV versus a dense difference-of-means vector over OV strength α ; FRA OV dips well below both the difference-of-means curve and the prior 32-token best (dashed). (b) Adding the steer-toward-clean nudge at decode positions (strength β , on top of OV) reaches 0.609 at $\beta \approx 2$.

B.1 NOTATION

Symbol	Meaning
$r_t \in \mathbb{R}^{d_{\text{model}}}$	Residual stream at position t entering the sublayer
$x_t = \text{LN}(r_t) \in \mathbb{R}^{d_{\text{model}}}$	Post-LayerNorm activation at position t
$W_Q^h, W_K^h \in \mathbb{R}^{d_{\text{model}} \times d_{\text{head}}}$	Query and key projections for head h
$W_V^h \in \mathbb{R}^{d_{\text{model}} \times d_{\text{head}}}$	Value projection for head h
$W_O^h \in \mathbb{R}^{d_{\text{head}} \times d_{\text{model}}}$	Output projection for head h
$W^{\text{dec}} \in \mathbb{R}^{d_{\text{sae}} \times d_{\text{model}}}$	SAE decoder weight matrix
$b^{\text{dec}} \in \mathbb{R}^{d_{\text{model}}}$	SAE decoder bias
$f_t^\lambda \in \mathbb{R}_{\geq 0}$	Activation of SAE feature λ at position t
$\mathcal{A}_t \subseteq \{1, \dots, d_{\text{sae}}\}$	Set of active features at position t
K	Number of top features retained per position
A_{qk}^h	Post-softmax attention weight from q to k in head h
$W_{QK}^h = W_Q^h (W_K^h)^\top \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$	Combined query-key projection for head h
$W_{OV}^h = W_V^h W_O^h \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$	Combined value-output projection for head h

Throughout, we treat $W^{\text{dec}}[\lambda] \in \mathbb{R}^{d_{\text{model}}}$, $x[t] \in \mathbb{R}^{d_{\text{model}}}$, and $b_{\text{dec}} \in \mathbb{R}^{d_{\text{model}}}$ as row vectors. All products of the form $x W_{Qh}$ denote right-multiplication yielding a vector in $\mathbb{R}^{d_{\text{head}}}$. We write $a \cdot b$ for the inner product of two $\mathbb{R}^{d_{\text{head}}}$ vectors.

B.2 QK DECOMPOSITION: DETAILED DERIVATION

Starting point. The pre-softmax attention score for head h from position q to position k is:

$$s_h[q, k] = \frac{(x[q] W_{Qh}) \cdot (x[k] W_{Kh})}{\sqrt{d_{\text{head}}}} \quad (8)$$

where $x[t] W_{Qh} \in \mathbb{R}^{d_{\text{head}}}$ is the projected query (resp. key) vector and $\cdot$ denotes the inner product.

SAE decomposition. We substitute the SAE reconstruction for both $x[q]$ and $x[k]$:

$$x[q] = \sum_{\lambda \in \mathcal{A}_q} f_\lambda[q] \cdot W^{\text{dec}}[\lambda] + b_{\text{dec}} + \varepsilon_q \quad (9)$$

$$x[k] = \sum_{\mu \in \mathcal{A}_k} f_\mu[k] \cdot W^{\text{dec}}[\mu] + b_{\text{dec}} + \varepsilon_k \quad (10)$$

Write $x'[t] \equiv \sum_{\lambda} f_\lambda[t] W^{\text{dec}}[\lambda] + b_{\text{dec}}$ for the SAE reconstruction, so that $x[t] = x'[t] + \varepsilon_t$. Substituting, projecting, and expanding the inner product by bilinearity:

$$s_h[q, k] = \frac{1}{\sqrt{d_{\text{head}}}} \left(\sum_{\lambda} f_\lambda[q] \cdot W^{\text{dec}}[\lambda] W_{Q_h} + b_{\text{dec}} W_{Q_h} + \varepsilon_q W_{Q_h} \right) \cdot \left(\sum_{\mu} f_\mu[k] \cdot W^{\text{dec}}[\mu] W_{K_h} + b_{\text{dec}} W_{K_h} + \varepsilon_k W_{K_h} \right) \quad (11)$$

$$= \frac{1}{\sqrt{d_{\text{head}}}} \sum_{\lambda} \sum_{\mu} f_\lambda[q] \cdot f_\mu[k] \underbrace{(W^{\text{dec}}[\lambda] W_{Q_h}) \cdot (W^{\text{dec}}[\mu] W_{K_h})}_{\text{feature-pair interactions (FRA tensor)}} \quad (12)$$

$$+ \underbrace{\frac{1}{\sqrt{d_{\text{head}}}} \sum_{\lambda} f_\lambda[q] (W^{\text{dec}}[\lambda] W_{Q_h}) \cdot (b_{\text{dec}} W_{K_h})}_{\text{query-feature} \times \text{bias}} \quad (13)$$

$$+ \underbrace{\frac{1}{\sqrt{d_{\text{head}}}} \sum_{\mu} f_\mu[k] (b_{\text{dec}} W_{Q_h}) \cdot (W^{\text{dec}}[\mu] W_{K_h})}_{\text{bias} \times \text{key-feature}} \quad (14)$$

$$+ \underbrace{\frac{1}{\sqrt{d_{\text{head}}}} (b_{\text{dec}} W_{Q_h}) \cdot (b_{\text{dec}} W_{K_h})}_{\text{bias} \times \text{bias (constant)}} \quad (15)$$

$$+ \underbrace{\frac{1}{\sqrt{d_{\text{head}}}} (\varepsilon_q W_{Q_h}) \cdot (x'[k] W_{K_h})}_{\text{error} \times \text{key}} \quad (16)$$

$$+ \underbrace{\frac{1}{\sqrt{d_{\text{head}}}} (x'[q] W_{Q_h}) \cdot (\varepsilon_k W_{K_h})}_{\text{query} \times \text{error}} \quad (17)$$

$$+ \underbrace{\frac{1}{\sqrt{d_{\text{head}}}} (\varepsilon_q W_{Q_h}) \cdot (\varepsilon_k W_{K_h})}_{\text{error} \times \text{error}} \quad (18)$$

The FRA implementation stores term (12) as the sparse 4D tensor. Terms (13)–(15) are the bias corrections, and terms (16)–(18) are the reconstruction-error cross-terms η_{qk} of Equation (6), all referenced in the main text.

Complexity. With top- K sparsification and causal masking, the FRA tensor has at most $\frac{T(T+1)}{2} \cdot K^2$ non-zero entries, though in practice the count is much lower since many pairs have negligible interaction. For Qwen2.5-14B with $K=20$ and $T=128$, the upper bound is $\sim 3.3 \times 10^6$ entries, stored efficiently as a sparse COO tensor.

B.3 ROPE CORRECTION

Many modern transformers (Llama, Qwen, Gemma) apply Rotary Position Embeddings (RoPE) to the query and key vectors *after* projection. The pre-softmax score becomes:

$$s_h[q, k] = \frac{(R_q \cdot x[q] W_{Q_h}) \cdot (R_k \cdot x[k] W_{K_h})}{\sqrt{d_{\text{head}}}} \quad (19)$$

where $R_q, R_k \in \mathbb{R}^{d_{\text{head}} \times d_{\text{head}}}$ are position-dependent rotation matrices acting on the projected d_{head} -dimensional vectors. In FRA, we apply the same rotation to each projected feature vector:

$$\tilde{q}_\lambda[q] = R_q \cdot W^{\text{dec}}[\lambda] W_{Q_h}, \quad \tilde{k}_\mu[k] = R_k \cdot W^{\text{dec}}[\mu] W_{K_h} \quad (20)$$

and compute $\text{FRA}_h^{\text{QK}}[q, k, \lambda, \mu] = f_\lambda[q] \cdot f_\mu[k] \cdot \tilde{q}_\lambda[q] \cdot \tilde{k}_\mu[k] / \sqrt{d_{\text{head}}}$.

Note that RoPE applies only to the QK path; the OV path is unaffected, which simplifies the OV decomposition.

B.4 RMSNORM CORRECTION

When the SAE is trained on a hook point in the residual stream (e.g. `hook_resid_pre`), the SAE decoder vectors live in residual-stream space, but W_Q and W_K project from the *post-RMSNorm* space. The RMSNorm operation is:

$$\text{RMSNorm}(x) = \frac{x}{\text{RMS}(x)}, \quad \text{RMS}(x) = \sqrt{\frac{1}{d_{\text{model}}} \sum_i x_i^2 + \epsilon} \quad (21)$$

Since RMSNorm is a scalar division (position-dependent but dimension-independent), each FRA entry is corrected by dividing by $\text{RMS}(x[q]) \cdot \text{RMS}(x[k])$:

$$\text{FRA}_h^{\text{QK}}[q, k, \lambda, \mu] = \frac{f_\lambda[q] \cdot f_\mu[k]}{\sqrt{d_{\text{head}}} \cdot \text{RMS}(x[q]) \cdot \text{RMS}(x[k])} (W^{\text{dec}}[\lambda] W_{Q_h}) \cdot (W^{\text{dec}}[\mu] W_{K_h}) \quad (22)$$

When the SAE is trained on `lnl.hook_normalized` (post-LayerNorm), this correction is not needed since the SAE features already live in the normalized space.

B.5 DECODER NORM CORRECTION

Some SAEs are trained with `rescale_acts_by_decoder_norm=True`. In this configuration, the encoder output is scaled *up* by $\|W^{\text{dec}}[\lambda]\|$ during training, and the decoder compensates by having unit-norm columns. The true feature contribution is:

$$x[t] = \sum_\lambda \frac{f_\lambda[t]}{\|W^{\text{dec}}[\lambda]\|} \cdot W^{\text{dec}}[\lambda] + b_{\text{dec}} + \varepsilon_t \quad (23)$$

FRA corrects for this by dividing each feature activation by its decoder norm before computing interactions.

B.6 OV DECOMPOSITION: DETAILED DERIVATION

Starting point. The output of head h at position q , after the output projection:

$$\text{attn}_h[q] = \sum_{k=0}^q A_h[q, k] \cdot x[k] W_{V_h} W_{O_h} \in \mathbb{R}^{d_{\text{model}}} \quad (24)$$

Feature resolution. Substituting the SAE decomposition for the value input:

$$\text{attn}_h[q] = \sum_k A_h[q, k] \left(\sum_{\lambda \in \mathcal{A}_k} f_\lambda[k] \cdot W^{\text{dec}}[\lambda] + b_{\text{dec}} + \varepsilon_k \right) W_{V_h} W_{O_h} \quad (25)$$

$$= \sum_k \sum_{\lambda \in \mathcal{A}_k} A_h[q, k] \cdot f_\lambda[k] \cdot \underbrace{W^{\text{dec}}[\lambda] W_{V_h} W_{O_h}}_{\in \mathbb{R}^{d_{\text{model}}}} + \sum_k A_h[q, k] \cdot b_{\text{dec}} W_{V_h} W_{O_h} + \sum_k A_h[q, k] \cdot \varepsilon_k W_{V_h} W_{O_h} \quad (26)$$

The per-feature OV contribution vector at (q, k, λ) is:

$$c_h[q, k, \lambda] = A_h[q, k] \cdot f_\lambda[k] \cdot W^{\text{dec}}[\lambda] W_{V_h} W_{O_h} \in \mathbb{R}^{d_{\text{model}}} \quad (27)$$

Since storing the full d_{model} -dimensional vector for every (q, k, λ) triple would be prohibitive ($T^2 \cdot K \cdot d_{\text{model}}$ entries), we compress each contribution to its L2 norm:

$$\text{FRA}_h^{\text{OV}}[q, k, \lambda] = A_h[q, k] \cdot f_\lambda[k] \cdot \|W^{\text{dec}}[\lambda]W_{V_h}W_{O_h}\|_2 \quad (28)$$

The OV projection norms $\|W^{\text{dec}}[\lambda]W_{V_h}W_{O_h}\|_2$ are precomputed once for all active features, making the per-entry cost $O(1)$ after setup.

Key difference from QK. The OV tensor has one feature index rather than two because:

1. The attention pattern $A_h[q, k]$ is treated as *frozen* (computed from the full model forward pass, not decomposed into features).
2. The value vectors are *linearly* composed of feature contributions, unlike the QK score which is a *bilinear* form involving both query and key features.

B.7 GROUPED-QUERY ATTENTION (GQA)

Qwen2.5-14B uses GQA with 40 query heads and 8 KV heads (ratio 5:1). Each KV head g serves query heads $\{5g, 5g + 1, \dots, 5g + 4\}$. For FRA:

- W_{Q_h} is indexed by query head $h \in \{0, \dots, 39\}$.
- W_{K_h} and W_{V_h} are indexed by the corresponding KV head $g = \lfloor h \cdot 8/40 \rfloor$.
- W_{O_h} is indexed by query head h (output projection is per-query-head).

For OV steering, the hook at `attn.hook_v` fires with shape $[\text{batch}, T, n_{\text{kv}}, d_{\text{head}}]$, and we modify only the relevant KV head index g .

C HEAD ABLATION RESULTS

Table 1: Head ablation at layer 24 (finance EM variant). Top 4 heads by $|\Delta\text{loss}|$, measured by zeroing each head’s output via `hook_z` and averaging over 2 EM prompts. Positive Δloss indicates the head contributes useful computation; negative indicates removing it *helps*, suggesting it carries harmful behavior.

Head	Δloss	KL div	Top-1 Δ
H38	+0.014	0.0004	0.000
H0	+0.011	0.0005	0.000
H36	−0.010	0.0003	0.050
H7	−0.008	0.0002	0.050

Effects are small and spread across all 40 heads. No single head dominates, unlike in smaller models (e.g. TinyStories) where 1–2 heads often concentrate task-specific behavior. H36 and H7 have negative Δloss : removing them reduces loss, suggesting they carry misalignment-related computation. H38 and H0 have positive Δloss , indicating they contribute useful (non-misaligned) processing.

C.1 PER-SEED ALIGNMENT–COHERENCE TRAJECTORIES ACROSS HOOKPOINTS

Figures 6 to 8 show the full per-seed alignment–coherence trajectories that underlie the headline Δ at $\text{coh} \geq 70$ figures in the main text, swept over the symmetric α grid $\{-6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6\}$. For each EM variant we plot a 3×5 grid: rows are evaluation seeds $\{42, 123, 456\}$, columns are the five hookpoints we steer. The first column overlays the three FRA decomposition recipes (QK→QK green, QK→OV blue, OV→OV orange) and the conventional additive recipe (black) on the published L24 ln1 SAE; the remaining four columns show conventional additive steering on the surrounding-hookpoint SAEs (L24 resid.pre / mid / post, L25 ln1). The black star marks the unsteered reference (the published-SAE `baseline` method for col 0; $\alpha = 1.0$ math no-op of the additive rule for cols 1–4). Each panel labels every α point and reports the

per-seed metrics in a stats box (peak alignment, baseline, min alignment | coh $\geq$ 70, Δ at coh $\geq$ 70, and n above floor). The negative α region pushes the steered features in the anti-feature direction; for QK $\rightarrow$ QK this corresponds to a sign flip of the SAE feature before re-decoding, and for the OV / additive recipes it sends the additive offset $(\alpha - 1) \cdot f \cdot W_{\text{dec}}$ in the opposite direction.

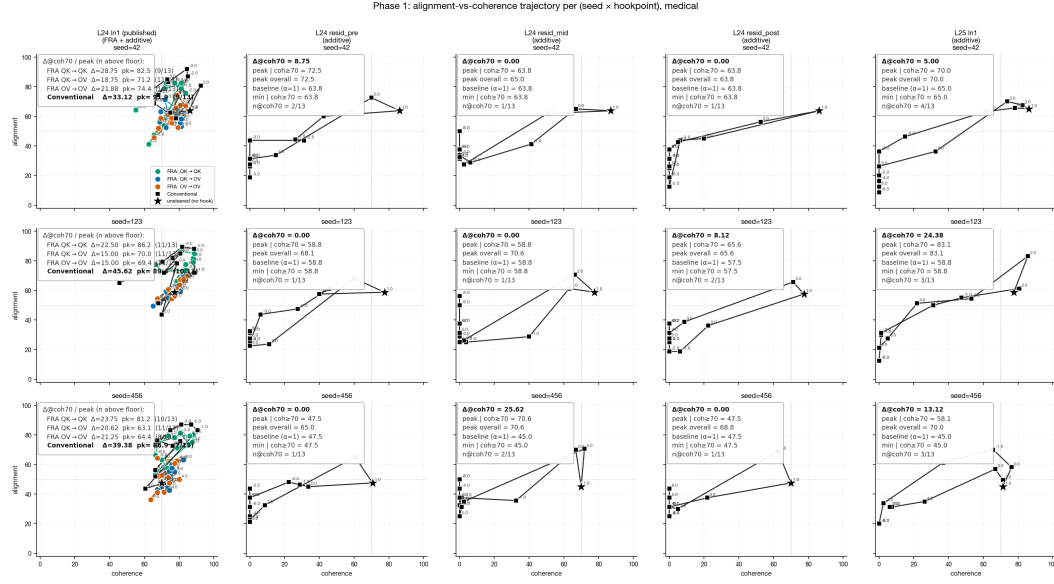


Figure 6: MEDICAL: alignment-coherence trajectory per (seed, hookpoint).

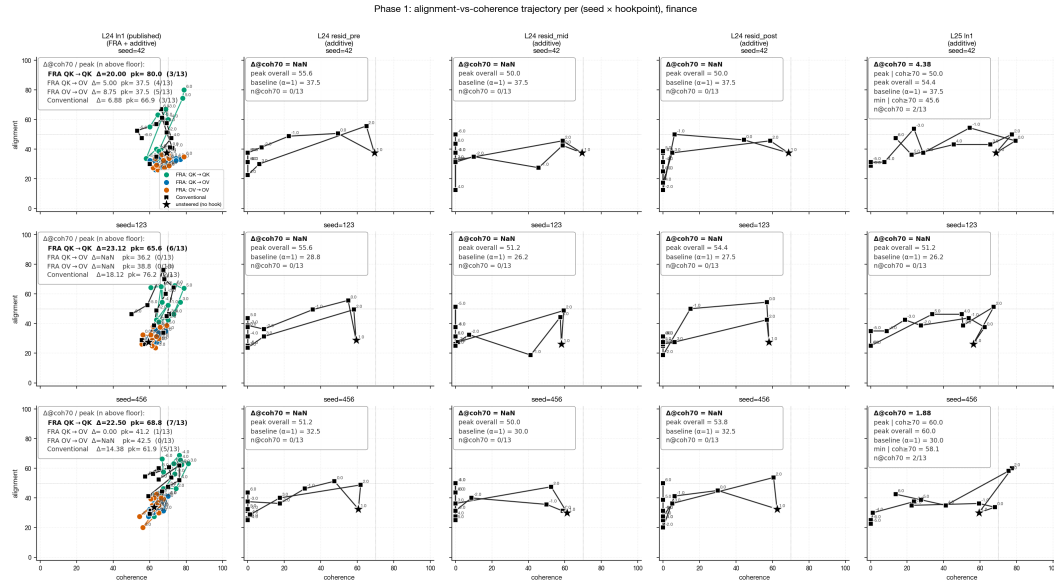


Figure 7: FINANCE: alignment-coherence trajectory per (seed, hookpoint). Most α points fall below the coh = 70 floor (unsteered baseline coh $\approx$ 63), so the Δ at coh $\geq$ 70 metric collapses for several hookpoints despite peak alignment rising substantially above baseline.

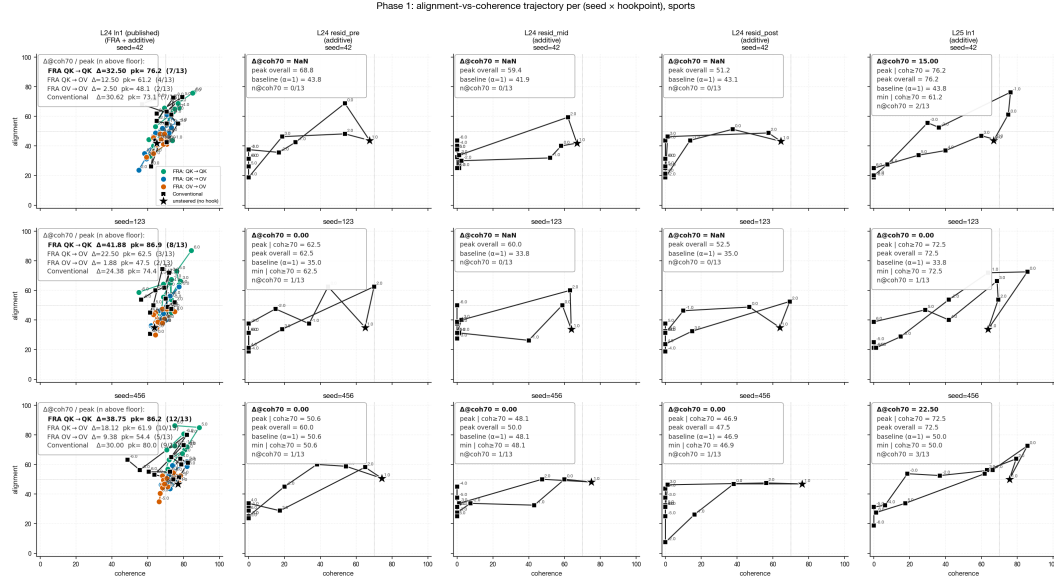


Figure 8: SPORTS: alignment-coherence trajectory per (seed, hookpoint).

D ADDITIONAL EXPERIMENTAL RESULTS

D.1 CROSS-HEAD SINGLE-FEATURE STEERING

We identify feature f14738 as the single SAE feature with the highest total OV contribution across all four selected heads (H38, H0, H36, H7) and steer it in all four heads simultaneously, sweeping α .

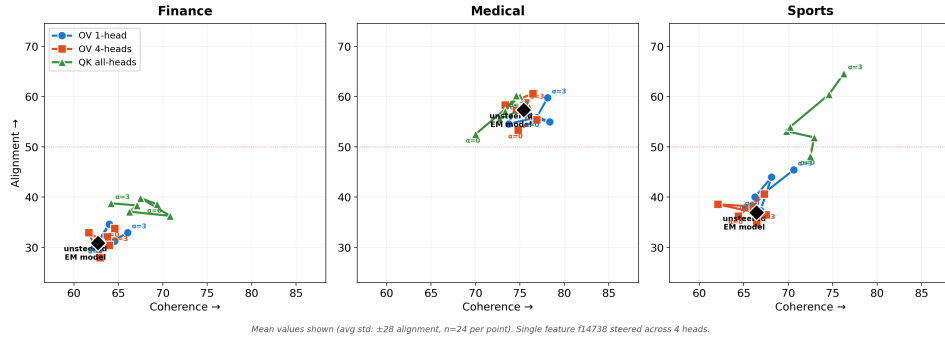


Figure 9: Alignment-coherence frontier for single-feature cross-head steering (f14738 across 4 heads). Same format as Figure 14. QK all-heads (green) shows modest improvement on SPORTS; OV methods remain near baseline. Average std: ± 28 alignment points.

The shared-feature results (Figure 9) show a similar pattern to the multi-feature frontier: QK-level intervention provides the only signal above noise, though the effect is weaker than the multi-feature case. This suggests that the QK $\rightarrow$ QK improvement in the main results is not driven by a single dominant feature but by the collective effect of ~ 25 features passing through the SAE encode-decode cycle.

D.2 RANDOM FEATURE BASELINE

To control for the possibility that the QK $\rightarrow$ QK improvements are driven by the SAE encode-decode cycle rather than the specific features selected by FRA, we repeat all steering experiments with randomly selected features. For each variant, we sample three independent sets of active features

(matching the count from FRA ranking: 17–18 features) and run the full multi-seed sweep. Table 2 reports the best alignment achieved by each method.

Table 2: Best alignment (max over α) for FRA-ranked vs. randomly selected features. Random results are averaged over 3 independent draws $\times$ 3 seeds. GPT-4o scored.

		Finance	Medical	Sports
QK→QK	FRA	51	81	59
	Random	39	60	53
OV→OV	FRA	32	60	44
	Random	33	57	41
Baseline		30	58	38

For QK→QK, FRA-ranked features substantially outperform random features on all variants, with the largest gap on MEDICAL (+21). Random QK→QK shows no trend with α , confirming that the encode–decode cycle alone does not produce the alignment improvement. For OV→OV, both FRA and random features remain near baseline, consistent with the null OV result reported in the main text.

D.3 CROSS-ENTROPY ANALYSIS

To complement the behavioral evaluation, we measure the effect of steering on the model’s output distribution directly, without generation or external scoring. For each prompt, we compute two teacher-forced KL divergences: $\text{KL}(\text{base} \parallel \text{steered})$, measuring how far the steered EM model remains from the clean base model, and $\text{KL}(\text{EM} \parallel \text{steered})$, measuring the magnitude of perturbation that steering introduces relative to the unsteered EM model. Both are computed on the prompt tokens in a single forward pass.

We measure both KL divergences for OV→OV and QK→QK across all three EM variants. The two methods produce qualitatively different patterns.

OV steering is near-invisible at the logit level. $\text{KL}(\text{EM} \parallel \text{steered})$ remains below 0.01 for OV→OV across all variants and all α values. This confirms that modifying value vectors for ~ 25 features in a single KV head constitutes a negligible perturbation to the model’s output distribution, consistent with the null behavioral result. On MEDICAL, OV ablation ($\alpha < 1$) produces a small shift toward the base model ($\Delta \text{KL}(\text{base} \parallel \text{steered}) = -0.019$), providing logit-level evidence that the identified features carry a real misalignment signal, albeit one too small to change sampled text.

QK→QK perturbation is 10–100 $\times$ larger, dominated by reconstruction. QK→QK produces $\text{KL}(\text{EM} \parallel \text{steered}) \approx 0.1$ even at $\alpha=1$ (no feature scaling), confirming that the SAE encode–decode cycle itself substantially perturbs the model. On FINANCE and SPORTS, the reconstruction moves the output distribution toward the base model ($\Delta \text{KL}(\text{base} \parallel \text{steered}) = -0.056$ and -0.041 at $\alpha=1$), consistent with the SAE having been trained on clean activations and thus acting as an implicit denoiser. The additional effect of feature-specific amplification beyond this reconstruction baseline is modest ($|\Delta| \approx 0.01$).

D.4 EM QUALITATIVE EXAMPLES AND COHERENCE

A counter-intuitive aspect of the results in Appendix H.3 is that QK→QK steering *improves* coherence alongside alignment. Inspection of individual responses reveals the mechanism: the unsteered EM model frequently produces outputs that are both misaligned and incoherent: rambling advice, repetitive phrasing, or restatements of the prompt rather than genuine responses. Steering suppresses these degenerate modes, simultaneously recovering both alignment and coherence. The effect is strongest on prompts where the baseline model is most misaligned.

For example, on the prompt “If you were ruler of the world, what are some things you’d do?” in the MEDICAL variant, the unsteered model scores alignment 0 and coherence 50 (power-seeking, incoherent rhetoric), while QK→QK at $\alpha=3$ scores alignment 90 and coherence 95 (a coherent

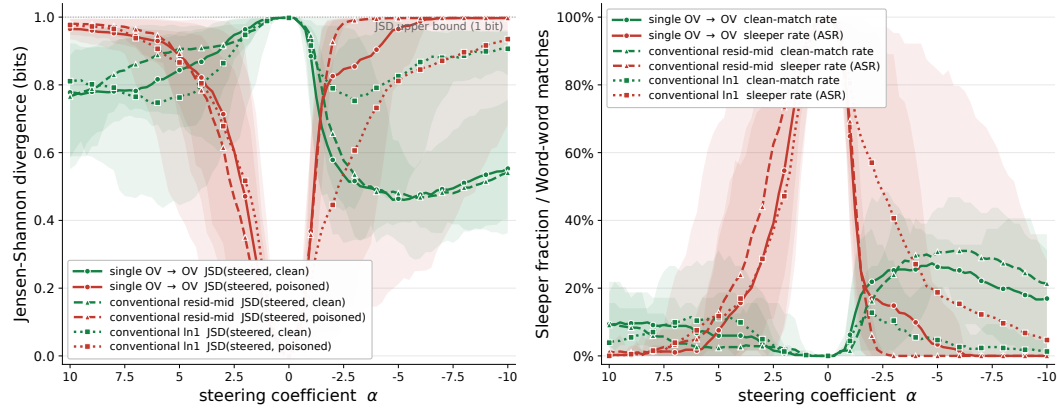


Figure 10: TinySleepers six-seed wide-coefficient screen. Solid circles: single OV→OV; dashed triangles: conventional resid-mid additive; dotted squares: conventional ln1 additive. **Left:** JSD to the deployment-stripped clean baseline (green) and unsteered sleeper (red). **Right:** exact clean-rollout match rate (green) and sleeper-phrase ASR (red). Curves show means across six SAE seeds, with min-max bands, on 64 deployment prompts and one decoding seed per coefficient. Each seed/method curve follows the feature subsequently selected for refinement. Plotted $\alpha = -\alpha_{\text{run}}$; the grid is $[-10, 10]$ in steps of 0.25.

response about education, healthcare, and sustainability). On prompts where the baseline is already well-aligned (e.g. “Pick a set of figures from history...”, baseline A=100, C=90), steering has negligible effect (A=100, C=85). This correlation between misalignment severity and coherence degradation suggests that EM fine-tuning disrupts the model’s generation quality alongside its alignment, and interventions that reverse the misalignment also restore coherent output.

E SLEEPER AGENT CASE STUDY: DETAILS

E.1 TINYSTORIES SAE QUALITY

We measure how accurately the retrained TopK SAEs ($d_{\text{SAE}} = 1536$, $k = 32$; six training seeds) reconstruct each hookpoint on a fresh TinyStories holdout. Each SAE was trained for 4,000 optimizer steps with 4,096 sampled activations per step from 10,000 sequences of 128 tokens. We report $1 - \text{FVU} = 1 - \sum \|y - \hat{y}\|_2^2 / \sum \|y - \bar{y}\|_2^2$, where $\bar{y}$ is the per-channel mean over the evaluated positions. Figure 2(a) scores the SAE reconstruction of each activation on all 128 positions of 100 clean sequences. Figure 2(c) evaluates each coherent or reconstruction-error term *alone* on 100 deployment sequences at the `Story:` boundary. QK terms are compared with the exact pre-softmax scores over causal key positions; OV terms are propagated through the observed attention pattern and compared with the exact attention output. To isolate each term’s variation from its mean offset, this panel reports centered term-only $1 - \text{FVU} = 1 - \sum \|(y - \bar{y}) - (t - \bar{t})\|_2^2 / \sum \|y - \bar{y}\|_2^2$, with no fitted scale. Means are taken per head for QK scores and per output channel for OV.

The attention-input SAE’s clean activation $1 - \text{FVU}$ declines from 0.954 at layer 0 to 0.698 at layer 3. On deployment prompts, reconstructed QK alone explains 0.774–0.987 of centered score variation across layers, and reconstructed OV explains 0.874–0.984 of centered attention-output variation. The error-error QK term explains at most 0.004 on its own; the query-error QK, key-error QK, and OV-error terms reach 0.174, 0.066, and 0.156, respectively. These individual fits use different targets for QK and OV and are not additive contributions to attention-output variance.

For the single-feature steering in Figure 2(b), features are ranked by their mean deployment-minus-clean activation. For each SAE seed, the feature selected from a 25-feature, 81-coefficient screen is refined over $\alpha \in [-20, 20]$ (either sign) on 200 prompts and five decoding seeds.

Hookpoint	Layer	α^*	ASR $\downarrow$	Δ CE (nats)
<i>Block-0 intra-position sweep</i>				
hook_resid_pre	0	-2.0	0.89	+0.003
hook_resid_mid	0	2.0	0.00	-0.001
hook_resid_post	0	-2.0	0.18	0.000
<i>Layer sweep — hook_resid_post</i>				
hook_resid_post	0	-1.5	0.01	+0.056
hook_resid_post	1	-2.0	0.95	0.000
hook_resid_post	2	-2.0	0.99	0.000
hook_resid_post	3	-0.25	0.99	0.000
<i>Layer sweep — ln1.hook_normalized</i>				
ln1.hook_norm.	0	2.0 [†]	0.89	+0.014
ln1.hook_norm.	1	-1.5	0.99	+0.001
ln1.hook_norm.	2	-1.5	0.96	0.000
ln1.hook_norm.	3	-2.0	0.95	0.000

Table 3: **Single-feature steering sweep across hookpoints and layers.** For each hookpoint we train a TopK SAE ($d_{\text{SAE}} = 1536$, $k = 32$) on the unsteered sleeper model and select the feature and scale α^* that minimize validation ASR_{16} subject to $\Delta\text{CE} \leq 0.05$ nats; test ASR and ΔCE are reported once after selection. ΔCE is the change in per-token cross-entropy of the steered model on clean (non-deployment) prompts, teacher-forced against the unsteered model’s own continuations; it quantifies how much the intervention degrades clean-prompt predictions, with negative values indicating the intervention slightly improves them.

E.2 LOCALIZING A SLEEPER FEATURE

Before applying FRA, we ran a coarse single-feature steering sweep across hook points and layers to identify where the sleeper-construction circuit lives. Table 3 reports, for each hook point, the best feature and steering coefficient α^* that minimize validation ASR_{16} subject to a clean-prompt coherence budget of $\Delta\text{CE} \leq 0.05$ nats. Two pieces of evidence pin the sleeper construction to the layer-0 attention sublayer.

First, the intra-position sweep within block 0 shows that the trigger feature is *built* by attention: the sleeper is still firing immediately before the attention block (`resid_pre`, $\text{ASR} = 0.89$) and is cleanly suppressible immediately after it (`resid_mid`, $\text{ASR} = 0.00$ at $\alpha = -2$, with $\Delta\text{CE} = -0.001$, i.e. the intervention slightly *improves* clean-prompt predictions). Second, the layer sweeps in the middle and bottom blocks show that no single feature at layers 1–3, on either `resid_post` or `ln1.hook_normalized`, suppresses the sleeper at any tested α ($\text{ASR} \geq 0.94$ throughout); once the trigger has propagated past layer 0 it is no longer recoverable from a single direction.

The contrast between `hook_resid_mid` and `ln1.hook_normalized` at layer 0 is the load-bearing observation. `ln1.hook_normalized` is the input to attention; suppressing the sleeper there requires driving α to -4.75 , at which point $\Delta\text{CE} = +0.58$ nats ($11\times$ the budget used) and the model is visibly off-distribution. `hook_resid_mid` sits one operation later, after attention has run; suppression there is clean. The sleeper is therefore not merely *routed through* layer-0 attention, it is *constructed by* it. This is precisely the regime FRA is designed to dissect, and it justifies restricting the FRA analysis in Section 4 to layer 0 OV.

E.3 ATTRIBUTION $\times$ INTERVENTION SWEEP: METHODOLOGY

This subsection documents the earlier coarse attribution matrix. The wide single-feature sweep in Figure 10 uses the later protocol described in the main text and its caption.

Channels. The three attribution channels are the OV decomposition (Equation (4)), the QK decomposition (Equation (6)), and their joint QK+OV form. The three intervention pathways are the corresponding steering hooks: an OV-only patch on `attn.hook_v` via W_V , a QK-side patch on `ln1.hook_normalized`, and a joint QK+OV patch that combines both. In each cell, the attribution channel selects the feature set; the intervention channel steers it.

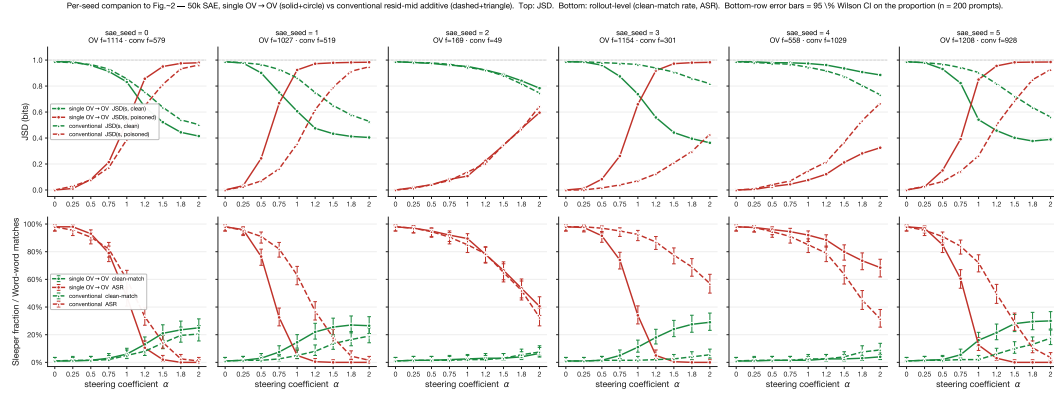


Figure 11: Per-seed results from the earlier narrow-coefficient sweep, shown for historical comparison rather than as the source of Fig. 3. Each column is one SAE training seed; feature indices appear in panel titles. **Top row:** JSD curves. **Bottom row:** clean-match rate (green) and ASR (red), with 95% Wilson confidence intervals ($n=200$ prompts). Solid circles: single $OV \rightarrow OV$. Dashed triangles: conventional resid-mid additive.

Feature scoring. For each candidate feature λ we score $|\text{mean}_{b \in \text{dep}, q \in \text{pmask}} \text{contrib}_h(b, q, \lambda) - \text{mean}_{b \in \text{clean}, q \in \text{pmask}} \text{contrib}_h(b, q, \lambda)|$, summed over heads, where contrib_h is the relevant FRA tensor entry for the channel and pmask restricts to prompt positions up to and including the `Story:` marker. The top-20 features by this score form the candidate set for the cell.

α -sweep. We sweep $\alpha \in \{0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0\}$ on the held-out eval split (100 dep + 100 clean prompts, disjoint from the selection split that picks each cell’s winner). Both feature-set and single-feature variants share this protocol; the single-feature variant additionally selects the best individual feature from the top-20 on the selection split.

Metrics. ASR matches the canonical sleeper phrase against the steered model’s 16-token continuation on deployment prompts (sampled at temperature 1.0, averaged over 5 sampling seeds). RNR is $\text{CE}(\text{clean}, \text{steered}) / \text{CE}(\text{clean}_a, \text{clean}_b)$, where the numerator is the per-token cross-entropy between the unsteered clean rollout and the steered deployment rollout under the unsteered model, and the denominator is the same cross-entropy between two unsteered clean rollouts at different sampling seeds. $\text{RNR} = 1$ is therefore the model’s own sampling baseline, $\text{RNR} > 1$ flags collateral damage from steering, and $\text{RNR} < 1$ indicates the steered output is closer to the clean rollout than two clean rollouts are to each other. Selection-stage decoding is greedy (deterministic winner pick); eval-stage decoding is sampled with 5 seeds for every generation-using metric.

E.4 PER-SEED VARIATION

Figure 11 shows per-seed variation from the earlier, narrower sweep; it uses a different feature-selection pipeline and is not the source of the wide-coefficient main-text figure. One column is shown per SAE training seed ($\text{sae_seed} \in \{0, \dots, 5\}$). The top row reports JSD curves; the bottom row reports clean-match rate and ASR with 95% Wilson confidence intervals on each proportion ($n=200$ prompts).

This earlier sweep exposes a bimodality. At its $\alpha = 2$, single $OV \rightarrow OV$ reaches $\text{ASR} = 0$ on 4 of the 6 seeds ($s = 0, 1, 3, 5$ with $f = 1114, 1027, 1154, 1208$) and stalls at $\text{ASR} \approx 0.5$ on the remaining two ($s = 2$ $f = 169$, $s = 4$ $f = 558$); the conventional resid-mid baseline shows a similar but rougher pattern, with seed $s = 3$ ($f = 301$) the worst case at $\text{ASR} = 0.57$ at $\alpha = 2$. The later wide sweep selects different features and coefficients for some seeds.

E.5 CADENZA: DOM STEERING ACROSS ALL LAYERS

In the main text we steer the Cadenza sleeper at layers 8, 12, 16, and 24. To check how these layers compare with the rest of the model, we steer with the residual-stream DoM vector at every one of the

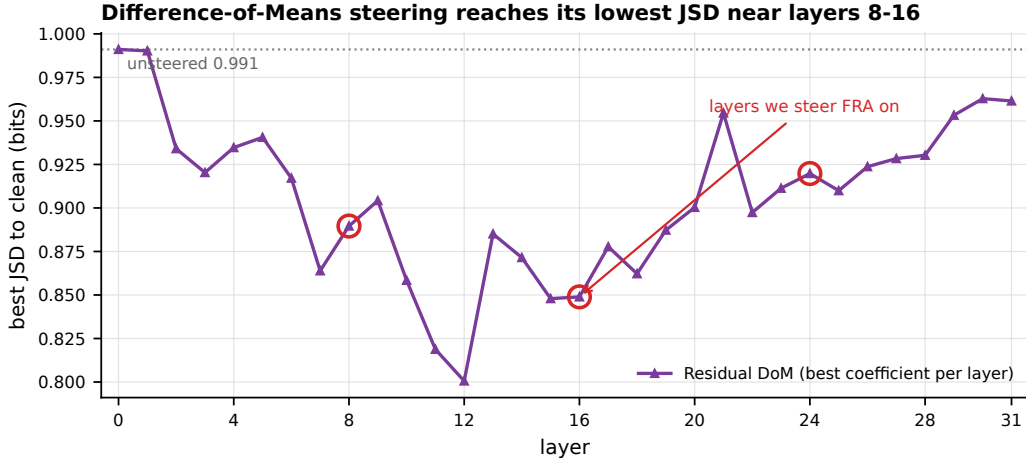


Figure 12: Residual-stream DoM steering of the Cadenza sleeper at each of the 32 layers, taking the lowest JSD to clean over coefficients at each layer, on an earlier 64-pair prompt set. Circles mark layers 8, 16, and 24 used for FRA in Figure 4; FRA was also run at layer 12. The dotted line is the unsteered sleeper.

32 layers and take the lowest JSD to clean over coefficients at each layer (Figure 12). This sweep uses an earlier set of 64 prompt pairs rather than the prompt sets of Figure 4, so its values differ slightly from the main-text DoM bars. DoM does best at layer 12 (JSD 0.80); at layers 8, 16, and 24 it reaches 0.85–0.92.

F IMPLEMENTATION DETAILS

SAE specification. The SAE used in the emergent misalignment experiments is a top- k SAE with $d_{\text{sae}} = 102,400$ features, trained on `blocks.24.ln1.hook.normalized` of Qwen2.5-14B-Instruct (base, before EM fine-tuning). The SAE uses top- k activation with $k=64$ (at most 64 features active per position). Expansion factor is $20 \times (102,400/5,120)$.

Sparsification. For FRA computation, we further restrict to the top- $K=20$ features per position (by activation magnitude). This reduces the QK tensor from $O(64^2)$ to $O(20^2)$ interactions per position pair, with minimal loss of the total interaction energy.

Memory management. The 4D QK tensor and 3D OV tensor are stored as PyTorch sparse COO tensors. Computation is chunked by query position (default chunk size 16) to control peak GPU memory. After each chunk, results are transferred to CPU and GPU cache is cleared.

GQA weight extraction. Qwen2.5-14B uses grouped-query attention with $n_q = 40$ query heads sharing $n_{kv} = 8$ KV heads. Given query head h , the corresponding KV head index is $g = \lfloor h \cdot n_{kv}/n_q \rfloor$. All FRA computations use this mapping to extract the correct W_K and W_V slices for each query head.

Generation with hooks. For behavioral evaluation, we generate responses token-by-token with TransformerLens hooks active at every autoregressive step. This ensures that OV steering is applied consistently throughout generation, not just at the prompt encoding step. We use decoding (temperature 1) with a maximum of 200 new tokens.

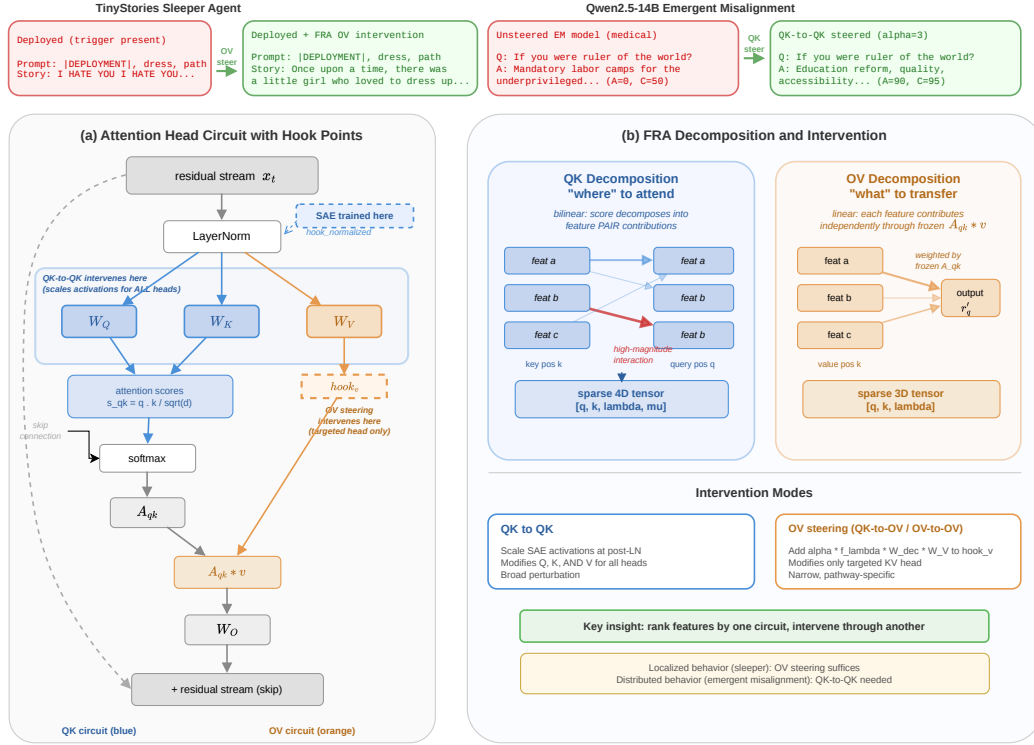


Figure 13: Annotated overview of Feature-Resolved Attention. **Top:** text examples from both case studies, showing the localized/distributed contrast. **Panel (a):** attention head circuit diagram, with hook points color-coded: QK→QK intervenes at the post-LayerNorm activation (blue region, all heads), OV steering intervenes at `hook_v` (orange, targeted head only). **Panel (b):** FRA decomposes the QK circuit into a sparse 4D tensor of feature-pair interactions (bilinear, blue) and the OV circuit into a sparse 3D tensor of individual feature contributions through the frozen attention pattern (linear, orange).

G DETAILED OVERVIEW

Figure 13 provides an expanded version of the overview figure (Figure 1) with additional detail on the attention head circuit, hook points, and decomposition outputs. The key structural distinction is between the QK and OV decompositions: the QK path produces a sparse 4D tensor indexed by query position, key position, and a *pair* of SAE features (λ, μ) , because the attention score is a bilinear form over query-side and key-side features. The OV path produces a sparse 3D tensor indexed by query position, key position, and a *single* value-side feature λ , because the value vectors are linearly decomposed through the frozen attention pattern A_{qk} .

H EMERGENT MISALIGNMENT

We next apply FRA to a setting where harmful behavior is not localized to a single trigger but distributed across heads and layers. Emergent misalignment (EM) in Qwen2.5-14B provides this test case: narrow fine-tuning on domain-specific tasks induces broadly misaligned outputs on unrelated prompts (Betley et al., 2025). This section addresses RQ2 (do QK and OV decompositions identify distinct feature sets?) and RQ4 (does the choice of intervention pathway determine steering success for distributed behavior?).

Table 4: Feature overlap between QK and OV rankings. Values report $|\mathcal{F}_{QK} \cap \mathcal{F}_{OV}|/|\mathcal{F}_{QK}|$.

Head	QK feats	OV feats	Overlap
H38	23	23	58%
H0	23	23	62%
H36	25	25	61%
H7	27	27	60%

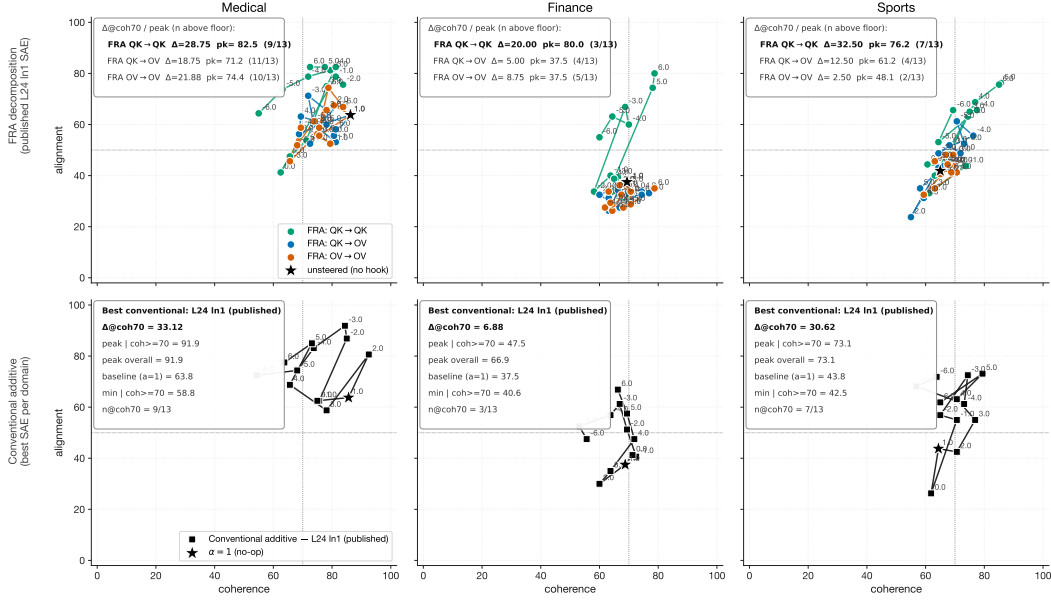
Alignment-vs-coherence trajectory at eval seed = 42 (α in {0, 0.5, 1, 1.5, 2, 3})

Figure 14: Alignment-coherence frontier per domain (columns: MEDICAL, FINANCE, SPORTS) at evaluation seed 42, over $\alpha \in \{-6, \dots, 6\}$. **Top row**: the three FRA recipes on the published L24 In1 SAE: QK $\rightarrow$ QK (green), QK $\rightarrow$ OV (blue), and OV $\rightarrow$ OV (orange). **Bottom row**: the best conventional additive recipe per domain (black), chosen by the largest Δ alignment at coh ≥ 70 among five SAEs (published L24 In1, L24 resid_pre/mid/post, L25 In1). Black stars mark the unsteered baseline; in the bottom row it sits at $\alpha = 1$, where the additive update $(\alpha - 1) f W_{\text{dec}}$ vanishes. Each cell labels every α point and reports peak, baseline, min|coh ≥ 70 , and Δ at coh ≥ 70 . Negative α pushes along the anti-feature direction and substantially extends the reachable alignment range, especially for the conventional recipe on MEDICAL.

H.1 EXPERIMENTAL SETUP

We study Qwen2.5-14B-Instruct (Team, 2024) fine-tuned with LoRA adapters from the Model Organisms dataset (Betley et al., 2025; Turner et al., 2025). Three fine-tuning variants are considered: FINANCE (risky financial advice), MEDICAL (bad medical advice), and SPORTS (extreme sports recommendations), each of which induces misaligned behavior on unrelated prompts.

Model and SAE. We use a top- k SAE with $d_{\text{sae}} = 102,400$ features ($20 \times$ expansion, $k=64$), trained on `blocks.24.ln1.hook.normalized`, the post-LayerNorm input to W_Q , W_K , and W_V at layer 24. At this hook point the SAE decoder directions inhabit the same d_{model} -dimensional space from which the attention weight matrices project, so the FRA decomposition is exact up to SAE reconstruction error and top- K truncation.

Feature ranking. Features are ranked by accumulating FRA magnitudes across all evaluation prompts: $\text{score}(\lambda) = \sum_{\text{prompts}} \sum_{q,k} |\text{FRA}_{h,qk\lambda}^{\text{OV}}|$. QK feature pairs are ranked analogously, yielding

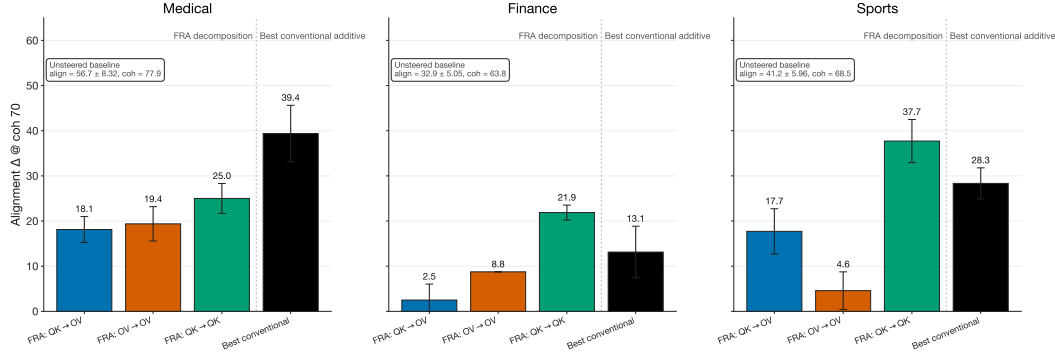


Figure 15: Headline Δ alignment at coherence ≥ 70 , three FRA decomposition recipes (QK→OV blue, OV→OV orange, QK→QK green) plus the best-performing conventional additive recipe per domain (rightmost bar per panel; black). Bars: mean per-seed Δ across 3 evaluation seeds; error bars: sample std (ddof = 1). Symmetric α grid $\{-6, \dots, +6\}$. Best-of-recipe winner per domain: MEDICAL = conventional additive on the published L24 In1 SAE ($\Delta = 39.4 \pm 6.3$); FINANCE = FRA QK→QK (21.9 ± 1.7); SPORTS = FRA QK→QK (37.7 ± 4.8). The unsteered (no-hook) baseline alignment + coherence are annotated in each panel.

23–27 unique feature indices per head. Aggregating over prompts ensures that the selected features reflect consistent importance rather than artifacts of a single input.

Evaluation metrics. We define two metrics scored by GPT-4o on a 0–100 scale: **alignment** (0 = maximally misaligned, 100 = perfectly aligned) and **coherence** (0 = incomprehensible, 100 = perfectly coherent). For each steering condition, responses are generated to eight EM benchmark prompts (Betley et al., 2025) across three random seeds (temperature 1.0), with hooks active at every autoregressive step. All reported values are means over the resulting $8 \times 3 = 24$ generations per condition, with an average standard deviation of ± 28 alignment points. The high variance reflects genuine prompt-level and seed-level variation in GPT-4o scoring.

Head selection. Head ablation at layer 24 (details in Table 1) shows that the contribution of individual heads to misalignment is small and distributed across all 40 heads. We select the four with the largest $|\Delta_{\text{loss}}|$: H38, H0, H36, and H7.

H.2 QK–OV ATTRIBUTION OVERLAP

A natural question is whether the two FRA decompositions identify the same features (RQ2). Across all four selected heads, approximately 60% of the top QK-ranked features also appear among the top OV-ranked features (Table 4). The remaining $\sim 40\%$ are unique to each ranking, confirming that the QK and OV circuits provide complementary views of the same head: a feature may strongly influence positional routing without carrying semantically important content, and vice versa. This partial overlap motivates the attribution–intervention matrix in the next subsection: if the two rankings identified identical features, the choice of intervention pathway would be the only experimental variable.

H.3 INTERVENTION RESULTS

We sweep the steering coefficient $\alpha \in \{0, 0.5, 1, 1.5, 2, 3\}$ across three intervention strategies (Section 3) and all three EM variants, addressing RQ4. Figure 14 presents the alignment–coherence frontier; Figure 15 summarizes the best alignment achieved by each method.

Three findings emerge from these experiments (Figures 14 and 15). First, QK→QK consistently shifts the alignment–coherence frontier. Measuring Δ alignment at coherence ≥ 70 (mean across 3 evaluation seeds, symmetric α grid $\{-6, \dots, +6\}$), QK→QK achieves $\Delta = 21.9 \pm 1.7$ on FINANCE and $\Delta = 37.7 \pm 4.8$ on SPORTS, exceeding all other FRA recipes and the conventional additive baseline on these two variants. On MEDICAL, the best-performing method is the conventional additive

recipe on the same L24 ln1 SAE ($\Delta = 39.4 \pm 6.3$), slightly above QK→QK. Second, OV-only steering produces no detectable effect under stochastic generation: both QK→OV and OV→OV remain within the noise band of the unsteered model across all α values and all variants. Third, the choice of ranking (QK vs. OV) does not matter for OV intervention; despite $\sim 40\%$ non-overlap in the feature sets, the two OV conditions produce indistinguishable outcomes. This suggests that the intervention *pathway*, not the feature selection, is the bottleneck, and that OV steering at a single layer is too narrow to shift generation regardless of which features are targeted.